FINAL PROJECT REPORT

Expense Tracker using Java

1. Introduction

The Expense Tracker is a console-based Java application developed to help users manage their daily expenses efficiently. The application enables users to record, view, search, calculate, and delete expense records through a simple menu-driven interface. The project was developed using Core Java concepts such as classes, objects, ArrayList, loops, conditional statements, and exception handling.

2. Objective

The main objectives of the project are:

- To record daily expenses systematically.
- To categorize expenses such as Food, Travel, Shopping, and Snacks.
- To calculate the total amount spent.
- To search expenses based on category.
- To delete unwanted expense records.
- To provide a simple and user-friendly expense management system.

3. Tools and Technologies Used

- Programming Language: Java
- IDE: Visual Studio Code
- JDK Version: JDK 17 (or your installed version)
- Concepts Used:
  - Classes and Objects
  - ArrayList
  - Scanner Class
  - Loops
  - Switch Case
  - Conditional Statements

4. Features of the Project

1. Add Expense

   - Users can enter date, category, description, and amount.

2. View Expenses

   - Displays all expense records stored in the application.

3. View Total Expense

   - Calculates and displays the total amount spent.

4. Search Expense by Category

   - Users can search expenses by category such as Food or Travel.

5. Delete Expense

   - Allows users to remove expense records based on description.

6. Exit

   - Safely exits the application.

5. Implementation

The project uses an Expense class to store expense details such as date, category, description, and amount. Multiple expense records are stored dynamically using ArrayList. The application uses a menu-driven approach where users can choose different operations using switch-case statements.

6. Challenges Faced

- Managing multiple expense records efficiently.
- Implementing search and delete functionality.
- Handling user input correctly.
- Organizing expense data using classes and objects.

7. Solutions Implemented

- Used ArrayList for dynamic storage.
- Implemented category-based search using loops.
- Used remove() method to delete records.
- Applied Object-Oriented Programming concepts for better code organization.

8. Learning Outcome

Through this project, I learned:

- Object-Oriented Programming concepts in Java.
- Usage of ArrayList and Scanner classes.
- Menu-driven application development.
- Searching and deleting records in Java.
- Managing data efficiently using classes and objects.

9. Conclusion

The Expense Tracker project was successfully developed using Java. The application provides an easy way to record and manage daily expenses. The implemented features such as add, view, search, calculate total, and delete expenses make the system efficient and user-friendly. This project enhanced my knowledge of Core Java and improved my programming skills.